Location, Not Shape: A Controlled Study of Anatomy-Guided Masking for Joint-Embedding Predictive Pretraining on Retinal OCT

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Abstract

Masked predictive pretraining asks a model to reconstruct the representation of hidden image regions. Which regions are hidden is a free design choice, and in medical imaging there is an intuitive answer: hide the tissue that carries the diagnosis. We test that intuition under tightly controlled conditions. Six masking policies are continued from a single shared I-JEPA checkpoint on 3D retinal OCT, differing *only* in how predictor targets are placed, and evaluated with a frozen linear-probe protocol for glaucoma classification (FairVision, N=3000). Guidance does help: restricting targets to retinal tissue improves AUC by $+0.0120$ over unguided masking at the matched epoch (95% CI $[+0.0068, +0.0173]$), and a band located by a first-order intensity statistic improves it by $+0.0109$ at epoch 100 ($[+0.0057, +0.0160]$, p<0.0001). All six pre-specified contrasts against the null, at all three epochs, exclude zero. But anatomical *precision* does not help. A policy that places 97% of masked patches on tissue, using a trained retinal segmentation model to shape the targets, is *indistinguishable from unguided masking* at the matched epoch ($+0.0013$, CI $[-0.0055, +0.0082]$), as is a coverage-constrained policy ($+0.0002$) — while the two policies that merely *place* rectangles on tissue gain an order of magnitude more. Direct measurement of mask geometry across 600 slices shows the fraction of anatomy hidden is uncorrelated with downstream AUC. The best policy consults no segmentation model at all. A subgroup audit over 27 probes finds the same worst-served group every time; gains reach every group and every disease stage, mild disease most of all ($+0.0137$), but the difficulty ordering is untouched: the severe-to-mild gap stays within 0.0133 of constant across every policy. Because each policy was pretrained once, we report these as single-run observations rather than an established ranking.

1 Introduction

Self-supervised learning has become the default route to representation quality in medical imaging, where expert annotation is expensive and pathology is rare [Azizi et al., 2021, Zhou et al., 2023, Qiu et al., 2024]. Joint-embedding predictive architectures such as I-JEPA [Assran et al., 2023] learn by predicting the *representation* of masked target blocks from a visible context block, avoiding pixel reconstruction and its bias toward high-frequency detail [He et al., 2022, Xie et al., 2022].

I-JEPA samples its targets geometrically: rectangles placed uniformly at random. In natural images this is defensible, since informative content is spread broadly across the frame. In a retinal OCT B-scan it is not obviously so. A large fraction of the image is dark vitreous and background; the diagnostic signal for glaucoma is concentrated in a thin band of retinal tissue. This motivates an intuitive hypothesis, one that several recent methods adopt [Li et al., 2022, 2024, Wang et al., 2025, Shi et al., 2022]:

> *Direct the predictive objective at clinically meaningful anatomy rather than at arbitrary image regions.*

We test this hypothesis directly and at several strengths. Rather than proposing one anatomy-aware method and reporting that it beats a baseline, we construct a *ladder* of masking policies of increasing anatomical specificity — from unguided rectangles, through rectangles constrained to lie on tissue, to ragged targets whose shape follows the segmented retina — and hold everything else fixed. All arms continue from one shared pretrained checkpoint, use identical optimiser, schedule, and effective batch size, and are evaluated under one frozen-probe protocol.

The result contradicts the intuition in an informative way. Guidance helps, but the benefit does not increase with anatomical precision — it disappears. The two policies whose targets track the segmented retina most faithfully are statistically indistinguishable from unguided masking at the matched epoch, while two policies that merely place ordinary rectangles on tissue gain an order of magnitude more. The best performer uses no segmentation model at all, locating a band by a per-column intensity centroid.

Contributions.

- A controlled six-arm study isolating masking policy in medical SSL, with a shared ancestor checkpoint and a single evaluation protocol (21 frozen probes).
- Evidence that guidance helps but anatomical precision does not, including the specific control — rectangles restricted to retinal tissue — that separates "avoid background" from "target anatomy".
- Direct measurement of mask geometry for every policy on 600 slices, showing that fraction-of-anatomy-hidden does not order downstream AUC, and an explicit accounting of the axes on which the anatomy arms are confounded.
- A subgroup audit and a 10^{-5} reproduction of the headline result from public weights on different hardware and precision.

2 Related work

Masked image modelling and JEPAs. Masked autoencoders reconstruct pixels [He et al., 2022, Xie et al., 2022]; BEiT [Bao et al., 2022] and iBOT [Zhou et al., 2022] predict discrete tokens; data2vec [Baevski et al., 2022] and I-JEPA [Assran et al., 2023] predict latent representations. Video and 3D extensions follow the same template [Bardes et al., 2024, Chen et al., 2023b, Taleb et al., 2020]. Recent analyses question how much of the benefit derives from the masking distribution itself [Balestriero and LeCun, 2025, He et al., 2025].

Informed masking. A substantial line of work argues that *what* is masked matters. ADIOS [Shi et al., 2022] learns an adversarial masking model; SemMAE [Li et al., 2022] uses learned semantic parts; AttMask [Kakogeorgiou et al., 2022] masks by attention; HardPatches [Wang et al., 2023a] and AutoMAE [Chen et al., 2023a] target difficulty. In the medical domain, AnatoMask [Li et al., 2024], AnatomyMAE [Ceballos Arroyo et al., 2026], MSMAE [Mao et al., 2023], and VasoMIM [Huang et al., 2026] explicitly steer masking toward anatomy, and Wang et al. [2025] argue directly for masking clinically relevant regions. Our study is a controlled test of that family's central premise on one task, and finds it does not hold in the direction usually assumed.

Evidence that informed masking is not uniformly better. The picture is less one-sided than the paragraph above suggests. The original masked-modelling papers already report that random sampling is hard to beat: MAE's ablation puts random-75 at 73.5 linear against block-75 at 63.9 [He et al., 2022], and SimMIM finds random beats its best block variant [Xie et al., 2022]. Among informed schemes, AttMask sits within 0.2 points of random [Kakogeorgiou et al., 2022], AutoMAE ties MAE under full fine-tuning [Chen et al., 2023a], HPM gains only when randomness is retained and *loses* when restricted to the hardest patches [Wang et al., 2023b], and attention-guided MAE underperforms MAE in the low-label regime [Sick et al., 2025]. Most directly, SemMAE's own part-quality ablation shows that *imperfect* semantic parts add nothing over no parts at all [Li et al., 2022], and Guo et al. [2025] reproduce SemMAE below random MAE under a common protocol, with the deficit largest under frozen evaluation — also our protocol. Chen et al. [2023a] name the mechanism a "patch selection dilemma": a slight foreground bias helps, but raising it too far starves the encoder of the evidence it must predict from. Conversely, the closest positive precedent for our best policy uses no segmentation model either: Lee et al. [2025] select foreground from normalised intensity alone and gain across five 3D medical benchmarks. Appendix G tabulates these.

We therefore do *not* claim to overturn the informed-masking literature. That random masking is a strong baseline, and semantic guidance wins only sometimes, is known and contested. What we add is a controlled measurement of where a *trained medical segmentation model* falls on that spectrum when everything else is held fixed, in a setting — 3D retinal OCT, joint-embedding prediction, frozen probe — where we could find no prior controlled comparison.

Retinal foundation models and OCT. RETFound [Zhou et al., 2023], VisionFM [Qiu et al., 2024], URFound [Yu et al., 2024], and OCTCube [Liu et al., 2026] pretrain on large retinal corpora; OCT-specific masked modelling is explored by Pissas et al. [2024]. MIRAGE [Morano et al., 2025] provides multimodal OCT segmentation and supplies the anatomical guidance used by several of our arms. Deep glaucoma detection from OCT is established [Maetschke et al., 2019, Ran et al., 2019, Noury et al., 2022].

Fairness in medical imaging. Performance disparities across demographic groups are widely documented [Seyyed-Kalantari et al., 2021, Larrazabal et al., 2020, Gichoya et al., 2022]. FairVision [Luo et al., 2024a] and Harvard-GF [Luo et al., 2024b] provide OCT data with demographic attributes; FairMedFM [Jin et al., 2024] and Shi et al. [2025] benchmark fairness for medical foundation models. We report a subgroup analysis as a secondary finding.

3 Method: a ladder of masking policies

3.1 Background

I-JEPA partitions the token grid of an image into a context block and M target blocks. A context encoder f_θ embeds the visible tokens; a predictor g_ϕ maps that embedding plus positional queries to predicted representations of the target tokens; an exponential-moving-average target encoder $f_{\bar{\theta}}$ supplies the regression targets. With a 16×16 token grid the design choice we vary is which of the 256 cells become targets.

3.2 The six policies

All policies produce $M{=}4$ target blocks and one context block under identical scale and aspect-ratio ranges. They differ only in placement and shape.

RANDOM (null). Stock I-JEPA multiblock. Four rectangles placed uniformly at random anywhere in the grid. No image content is consulted.

ENVELOPE (random-within-retina). Identical rectangles — same shape, size and count as RANDOM — but rejection-sampled onto the retinal envelope predicted by MIRAGE [Morano et al., 2025]. Only *location* changes. This is the control that separates "stop wasting targets on background" from "target anatomy specifically".

INTENSITY (segmentation-free). A band whose vertical position is located per slice by a per-column intensity-weighted row centroid, smoothed across columns. It adapts to the retina's position and curvature using a first-order intensity statistic, and consults **no segmentation model**. It uses no ground-truth annotation of any kind. (Our released code and checkpoints label this policy `oracle`, a historical name from when it was intended as an upper bound. We rename it here because it is neither an oracle nor an upper bound: it reads only the input image.)

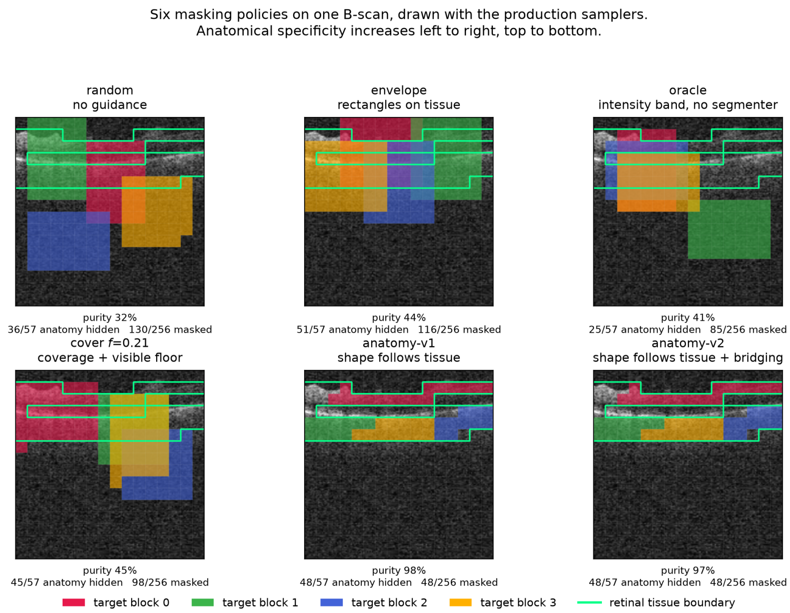

Figure 1: The six masking policies on one identical B-scan, drawn with the production samplers. Each predictor target block has its own colour; the green contour is the anatomy reference. Anatomical specificity increases along the ladder: ENVELOPE and COVER place rectangles on tissue, while the anatomy arms grow targets that trace the retina almost exactly. Downstream AUC does not increase with that specificity (Table 1). Figure 4 repeats this over five B-scans spanning the volume.

ANATOMY-V1 / ANATOMY-V2. Ragged connected components grown on the MIRAGE score map, so target *shape* follows tissue rather than being rectangular. v2 additionally bridges diagonal adjacencies.

COVER f. Targets chosen greedily to cover the anatomy subject to a hard floor leaving a fraction f of tissue visible in the context. We pretrain $f=0.21$.

Figure 1 shows all six on the same B-scans.

3.3 Hypotheses

The ladder yields three testable statements:

H1 *Avoiding background helps.* ENVELOPE $>$ RANDOM.

H2 *Anatomical precision helps further.* anatomy arms $>$ ENVELOPE.

H3 *Any benefit is due to anatomical targeting, not to incidental changes in mask area, target count, or context size.*

4 Experimental setup

Data. FairVision [Luo et al., 2024a] glaucoma OCT volumes. Each volume is a 3D scan resampled to 100 B-scans of 256×256 after transform. The label is a binary glaucoma diagnosis. Splits are fixed and patient-disjoint as released; downstream evaluation uses a held-out test split of $N=3000$ volumes (1466 positive / 1534 negative), seed 42, fixed loader order. No test volume is seen during pretraining, probe fitting, or model selection: the probe head is selected on a separate validation split. Labels are byte-identical across arms, so any two arms are paired-comparable on the same cases. The dataset carries self-reported demographic attributes, which we use only for the subgroup analysis in Section 5.3 and never as model inputs.

Table 1: Frozen MeanPool probe, FairVision test (N=3000, 1466 positive). All arms continue from one epoch-25 ancestor (AUC 0.8487). Δ is versus RANDOM at the *matched* epoch. The upper block is precision-matched (fp16 throughout) and carries every headline claim. The lower block is fp32 and its Δ is taken against an fp32 null re-probed for that comparison, so those rows are matched on epoch and precision too.

		epoch 50		epoch 75		epoch 100	
policy	guidance signal	AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	none (null)	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	segmenter (location only)	0.8761	**+0.0120**	0.8803	**+0.0080**	0.8807	**+0.0062**
INTENSITY	intensity centroid, no segmenter	0.8740	**+0.0099**	0.8836	**+0.0113**	**0.8855**	**+0.0109**
ANATOMY-V2	segmenter (target shape)	0.8654	+0.0013	—	—	—	—
COVER f=.21	segmenter (coverage)	0.8643	+0.0002	—	—	pending†	pending†

Pretraining. Patch-level I-JEPA, ViT-Base, 256×256 crops, patch size 16, predictor depth 6, EMA target encoder. **Every arm resumes from the same epoch-25 checkpoint** and continues with identical optimiser, learning-rate and weight-decay schedules, and effective batch size (512 for all arms). Guidance is ramped linearly from epoch 25 to 30 and held at full strength thereafter. The *only* difference between arms is the mask sampler.

Arms differ in how far they were carried, and we do not pretend otherwise. RANDOM, INTENSITY and ENVELOPE reach epoch 100. ANATOMY-V2 has valid probes to epoch 50 (later checkpoints follow a target-precision splice and are excluded, Appendix D). ANATOMY-V1 has only epoch 30. COVER f=0.21 was deliberately halted at epoch 73. Epoch 50 is therefore the only epoch at which five policies are simultaneously available, and it carries every cross-policy comparison that involves the anatomy or coverage arms.

Evaluation. The encoder is frozen. Per-slice features are mean-pooled over the volume, followed by LayerNorm and a linear head. The head is selected by validation AUC and reported on the held-out test split. We report paired bootstrap confidence intervals over test volumes (10,000 resamples, stratified by class, the same resampled index set applied to every arm) and DeLong tests for correlated ROC curves [DeLong et al., 1988], both of which are valid here because all arms are evaluated on the same cases in the same order. Each test volume is a distinct subject, so case-level resampling is already subject-level and no cluster correction is required [Obuchowski, 1997, Rutter, 2000]. We follow Maier-Hein et al. [2024] in reporting the operating-point metrics alongside AUC rather than AUC alone (Appendix A).

Numerical precision, and how we handle it. The probe harness defaults to mixed precision. The RANDOM, INTENSITY and ENVELOPE long-horizon probes were fitted under that default (fp16); the ANATOMY-V1, ANATOMY-V2, COVER and ancestor probes were fitted with autocast explicitly disabled (fp32). We therefore do *not* claim a single protocol across every probe in the study. Instead we partition the comparisons: all headline contrasts in Table 1 are drawn from arms sharing both epoch *and* precision, and any contrast that would cross the precision boundary is marked as such and excluded from headline claims. Section 5.4 reports a direct measurement of how large the precision effect actually is.

5 Results

What these numbers can and cannot establish. Each policy was pretrained exactly once, so policy is perfectly confounded with one stochastic optimisation path after the fork. The intervals below quantify sampling error over test subjects for a *fixed* pair of score vectors, not seed-to-seed variance of pretraining. Read every comparison here as a controlled single-run observation, not an established ranking over retrainings (Section 6).

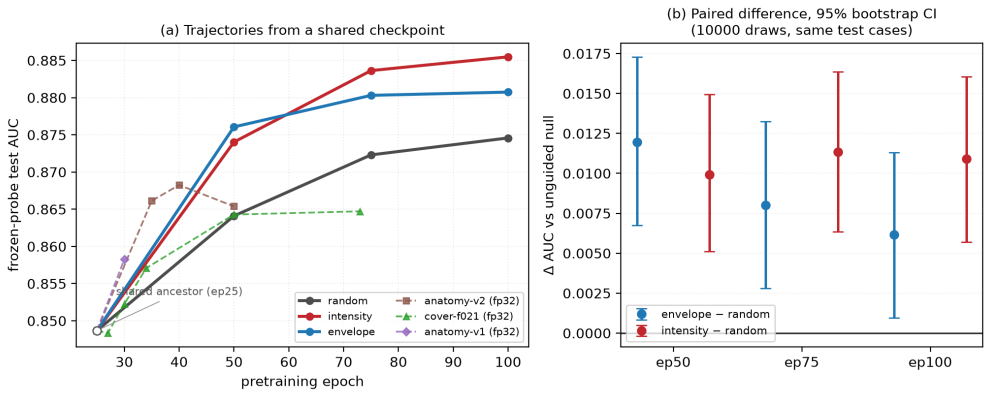

Figure 2: **(a)** Frozen-probe test AUC against pretraining epoch. All arms fork from the same epoch-25 checkpoint (open circle) and differ only in masking policy. Solid lines are the precision-matched fp16 family; dashed lines are fp32 arms and must not be compared vertically against the solid lines. **(b)** The actual inference: paired differences against the unguided null on identical test cases, with 10,000-resample bootstrap intervals. All six intervals exclude zero. We plot paired differences rather than per-arm error bars because marginal intervals carry between-case variance that cancels in a paired comparison, and would understate the evidence.

5.1 H1 holds, H2 fails

Table 1 gives the main comparison, Table 7 in Appendix H the full paired inference for all nine precision-matched contrasts, and Figure 2 the trajectories. All six contrasts against the null survive Benjamini–Hochberg correction over that family of nine ($q \leq 0.0299$ throughout). **H1 holds**: constraining the same rectangles to lie on retinal tissue (ENVELOPE) beats unguided masking by $+0.0120$ at epoch 50 (CI $[+0.0068, +0.0173]$) and by $+0.0062$ at epoch 100 (CI $[+0.0010, +0.0113]$). Avoiding background is worth something, at every epoch we measured, with an interval that excludes zero.

H2 fails. The arms that use the segmentation model most aggressively — shaping targets to trace the retina, or maximising anatomy coverage under a floor — do not improve on the null. At epoch 50, the only epoch at which all five arms have a probe, ANATOMY-V2 gains $+0.0013$ over RANDOM and COVER gains $+0.0002$, against $+0.0120$ for ENVELOPE and $+0.0099$ for INTENSITY. The policies that merely *place* rectangles on tissue gain roughly an order of magnitude more than the policies that shape targets to match it. Increasing anatomical fidelity past "rectangles on tissue" does not add benefit.

We are careful not to overstate this. ANATOMY-V2 is not *worse* than the null at the matched epoch; it is statistically indistinguishable from it (CI $[-0.0055, +0.0082]$, $p = 0.7153$), as is COVER (CI $[-0.0050, +0.0053]$, $p = 0.9513$). Both are measured against an fp32 null re-probed specifically for this comparison, so they are matched on epoch *and* precision. What the data rule out is not that these policies are harmful, but the expected *monotone improvement with anatomical precision*: their intervals are centred near zero and comfortably contain it, while the two cruder policies produce intervals that exclude it.

The strongest policy is the one that uses no segmentation model. INTENSITY gives $+0.0109$ over the null at epoch 100 (95% CI $[+0.0057, +0.0160]$, p<0.0001, q =<0.0001), with a margin that is stable across training ($+0.0099$, $+0.0113$, $+0.0109$). It exceeds the segmenter-guided ENVELOPE arm at epoch 100 by $+0.0047$ (CI $[+0.0004, +0.0091]$, $p = 0.0294$), though the two are indistinguishable at epochs 50 and 75. Notably, ENVELOPE's margin over the null *decays* with training ($+0.0120 \rightarrow +0.0080 \rightarrow +0.0062$) while INTENSITY's does not.

5.2 H3: it is not anatomical targeting

To test H3 we ran each production sampler on 600 held-out slices and measured what it actually does (Table 2, Figure 3). Three observations follow.

Table 2: Measured mask geometry, 600 held-out slices, production samplers. "purity" is the fraction of masked cells lying on tissue; "loss slots" is the number of predictor targets contributing to the loss. All values measured. AUC is quoted at the *matched* epoch 50, the only epoch at which all five policies have a probe, so that the geometry-to-AUC association is not read across mixed epochs. All five AUCs are at epoch 50.

policy	anatomy hidden	purity	mask ratio	context kept	loss slots	AUC @ep50
RANDOM	52.2%	31.6%	43.7%	42.1%	157.7	0.8641
INTENSITY	62.2%	41.1%	40.0%	45.6%	158.4	0.8740
ENVELOPE	76.9%	43.5%	46.4%	40.5%	159.9	**0.8761**
COVER	74.1%	45.3%	43.3%	43.5%	160.0	0.8643
ANATOMY-V2	80.3%	**97.3%**	21.4%	67.9%	64.0	0.8654

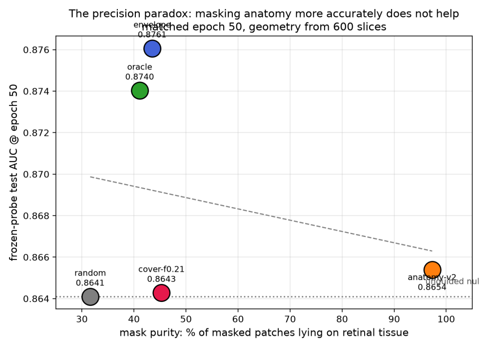

Figure 3: Downstream AUC at matched epoch 50 against mask purity, the fraction of masked patches lying on retinal tissue. The policy whose masks are 97% on-tissue does not separate from the null, while policies at 41–44% purity gain the most. The ordering does not support the intuition that more accurate anatomical masking yields better representations. With four to five arms this is descriptive, not inferential; the same result arranged as a specificity ladder is in Appendix I.

First, **anatomical targeting does not order the results**. Over the four rectangle arms, the Spearman correlation between fraction-of-anatomy-hidden and AUC is exactly 0.00: knowing how much anatomy a policy hides tells you nothing about its downstream AUC. Mask *purity* — the share of masked patches lying on tissue — trends negatively (-0.40). Neither is significant at $n{=}4$ arms, and we present them as descriptive rather than inferential; but neither shows the positive relationship the anatomy-guidance hypothesis predicts. The anatomy arm places 97.3% of masked cells on tissue and does not separate from the null; INTENSITY places 41.1% on tissue and gives the largest gain in the study.

Second, **the anatomy arms are confounded on several axes simultaneously**, and we state this rather than attributing their deficit to shape. Because their targets are compact tissue blobs rather than large rectangles, they hide only 21.4% of the grid against 40–46% for the rectangle arms, leave 67.9% of tokens visible as context against 40–46%, and supply 64 predictor loss slots against 158–160. Any of these could account for their deficit without invoking target shape at all. We therefore do *not* claim that irregular target shape is harmful; that comparison is not identified by this design. Figure 12 makes the four-way divergence explicit.

Third, INTENSITY is the most *spatially consistent* policy we measured. Its per-cell hit map is the most concentrated of any arm (Gini 0.400 versus 0.316–0.338 for the other rectangle arms): it repeatedly asks the encoder to represent the same region of the image. This suggests a mechanism other than anatomical correctness — consistency of the predictive task — which we return to in Section 6.

5.3 Subgroup analysis

Joining saved test predictions to the released FairVision metadata in deterministic loader order (the join is validated by exact label reconstruction: the label vector rebuilt from the sorted file order matches the stored labels for all 3000 cases in every probe), we audit seven stratifications across 27 probes spanning aggregate AUC 0.8483 to 0.8855. Probes that the exclusion rules of Appendix D remove are removed here too.

The ordering of subgroups never changes. The worst-performing group is the same in every one of the 27 probes for sex (female), race (black), ethnicity (hispanic) and disease severity (mild (-6,-2]). No masking policy reorders them. The probes share a test set, an epoch-25 ancestor and a probe seed, and several are different epochs of one branch, so they are *not* independent; we report this as a descriptive regularity and attach no p-value. What it shows is that the disparity is not introduced by any policy we tried.

Every group improves; no gap reliably moves. A widening max–min gap is easy to misread as harm. It is not. At epoch 100 the strongest policy improves *every* race stratum over the null — black $0.8325 \rightarrow 0.8472$ ($+0.0147$, a larger gain than white's $0.8741 \rightarrow 0.8853$) — and the max–min gap widens only because the already-best asian subgroup gains most. Nor is there a defensible gap *trend*: the race gap correlates with aggregate AUC at $\rho = +0.487$ ($p = 0.0100$) across checkpoints, but this fails Benjamini–Hochberg correction over the seven attributes tested ($q = 0.0232$) and vanishes when pseudo-replication is removed by averaging within each of the 11 branches ($\rho = +0.518$, $p = 0.1025$). The only gaps that survive both are sex and severity, and both *narrow*. We do not claim that better masking harms any group, and we do not claim it helps equity either.

Early disease improves, but stays hard. Scoring each severity stratum against the shared pool of all negatives, the strongest policy improves mild disease from 0.8164 to 0.8301 ($+0.0137$ at matched epoch 100), moderate by $+0.0102$ and severe by $+0.0063$: mild disease improves *most*. What does not move is the difficulty ordering. The severe-to-mild gap is 0.1361 for the null and 0.1286 for the best policy, and stays within 0.0133 of constant across all 27 probes while aggregate AUC spans 0.8483–0.8855. Target placement lifts the whole curve; it does not change its shape. Appendix C gives all seven stratifications. These are secondary, descriptive analyses; no policy here was designed as a fairness intervention and none should be presented as one.

5.4 Numerical precision is not the effect

Because the long-horizon arms and the anatomy arms were probed at different autocast settings (Section 4), we measured the precision effect rather than assuming it away. Re-encoding the epoch-100 INTENSITY test split at fp32 reproduces the reported AUC to 9.8×10^{-6}, and a full re-fit of the entire probe pipeline at fp32 shifts every long-horizon arm by less than 2×10^{-4}. Both are two to three orders of magnitude below the effects in Table 1. Appendix I gives the numbers.

5.5 Reproducibility

The epoch-100 INTENSITY encoder and its trained probe head will be released, together with the stored per-case predictions for every probe in this paper. Links are withheld here for anonymity. Every number in this paper is emitted by a single script from those stored predictions, so tables, figures and prose cannot disagree; that script is released with the code.

6 Discussion and limitations

What the evidence supports. Guidance helps: restricting targets to tissue is worth $+0.0120$ at the matched epoch, and a consistent intensity-located band is worth $+0.0109$ at epoch 100, with paired intervals excluding zero at every epoch measured. A trained segmentation model is not required to obtain this benefit, and using one more aggressively — to shape targets or to maximise anatomy coverage — does not add to it. The practical reading, bounded by the caveats below, is that for *this* task the segmentation stage bought us nothing we could measure, and a one-line intensity statistic recovered the whole benefit at a fraction of the engineering cost. We would not generalise that to tasks where anatomy is the output — layer segmentation, lesion localisation, progression — where segmentation-aware pretraining may well matter.

What it does not support. We cannot claim anatomy-shaped masking is *harmful*: at the matched epoch it sits $+0.0013$ from the null with an interval ($[-0.0055, +0.0082]$) that comfortably contains zero, so it is indistinguishable rather than worse. Nor can we claim target *shape* is the operative variable, for the reasons in Table 2. Nor can we claim the mechanism behind INTENSITY's advantage. Two explanations remain live: consistency of location (the encoder is repeatedly forced to represent the same region), and masking ratio or task difficulty. Distinguishing them requires an arm that randomises the band's vertical position while holding area and context fixed, which we did not run.

One pretraining run per policy. The paired bootstrap quantifies sampling error on a fixed test set; it does not quantify seed-to-seed variance of pretraining. With $n{=}1$ continuation per arm the *ranking* is not statistically established even where individual pairwise differences are significant on this test set. Bouthillier et al. [2021] show that single-seed comparisons routinely mistake optimisation noise for method effects. We can bound one component: with five probe seeds on two fixed encoders the probe-seed standard deviation is 0.0003 and 0.0018, and the arms stay fully separated across all five, so probe noise is an order of magnitude below the effects in Table 1. Those are *technical* replicates and do not estimate pretraining noise, which would need at least three paired continuations from the shared checkpoint with the continuation as the unit of replication. That is the single most valuable follow-up and we do not claim its result in advance.

Scope, coverage and an incomplete arm. All results are glaucoma classification from FairVision OCT; generality to other modalities, diseases, or segmentation-quality regimes is untested, and the anatomy arms depend on MIRAGE's prediction quality on this data, so a better segmenter might change the conclusion. Only $f{=}0.21$ was pretrained for COVER, so we report no dose-response over the visible-tissue floor and make no claim that any floor is optimal. That arm also has probes only at epochs 27, 30, 34, 50 and 73: it was deliberately halted before epoch 100, its later probes are marked pending rather than reported, and we draw no conclusion about its endpoint. Its epoch-50 value is included because it is a matched-epoch comparison against every other arm. Finally, one dataset and one test split were reused throughout this programme's history, and policies, checkpoints and analyses were chosen after repeated inspection of that split; the intervals we report are therefore best read as descriptive rather than as confirmatory inference.

Broader impact and ethics. This study uses a public, de-identified retinal dataset under its release terms and involves no new data collection or human subjects. No model reported here is clinically validated or intended for deployment. Appendix F gives the full statement.

7 Conclusion

We tested the intuition that masked predictive pretraining in medical imaging should target clinically meaningful anatomy. Under matched compute, schedule, and a shared ancestor checkpoint, guidance helped — but anatomical precision did not. Restricting targets to retinal tissue improved AUC over unguided masking by $+0.0120$ at the matched epoch, while shaping targets to trace the segmented retina, or maximising anatomy coverage, left performance indistinguishable from not guiding at all; and the best policy consulted no segmentation model. Mask geometry is uncorrelated with how much anatomy is hidden, and a subgroup audit over 27 probes shows the gains do not reach the groups or the disease stage that need them most. We suggest the operative variable is the consistency of the predictive task rather than its anatomical correctness, and will release weights and the reproduction protocol.

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A All frozen probes

Table 3 lists every frozen MeanPool probe in the study.

Table 3: All frozen probes. Protocol identical throughout: FairVision test N=3000, seed 42, frozen encoder, MeanPool + LayerNorm + Linear.

policy	epoch	test AUC	target precision
ancestor (RANDOM @ep25)	25	0.848680	fp32
RANDOM	50	0.864097	fp32
RANDOM	75	0.872302	fp32
RANDOM	100	0.874581	fp32
INTENSITY	50	0.874030	fp32
INTENSITY	75	0.883636	fp32
INTENSITY	100	**0.885485**	fp32
ENVELOPE	30	0.853917	fp32
ENVELOPE	50	0.876064	fp32
ENVELOPE	75	0.880307	fp32
ENVELOPE	100	0.880743	fp32
ANATOMY-V1	30	0.858274	fp32
ANATOMY-V2	35	0.866129	fp32
ANATOMY-V2	40	0.868251	fp32
ANATOMY-V2	50	0.865386	fp32
ANATOMY-V2	75	0.862492	fp16 (confounded)
ANATOMY-V2	92	0.860364	fp16 (confounded)
COVER f=.21	27	0.848347	fp32
COVER f=.21	30	0.852249	fp32
COVER f=.21	34	0.857083	fp32
COVER f=.21	50	0.864281	fp32

B Masking policies across the volume

Figure 1 shows one B-scan for legibility. Figure 4 repeats the same rendering on five B-scans sampled across the depth of a volume, so that the stability of each policy under changing retinal geometry can be inspected. The samplers, the colour assignment, and the anatomy contour are identical to Figure 1.

C Subgroup and severity analysis

All numbers below come from saved test predictions joined to FairVision's released metadata in deterministic loader order; the join is validated by exact label reconstruction (3000/3000 labels recovered in every probe). The 27 probes used here are exactly those the inventory marks valid: the four retracted coverage probes of Appendix D and the two precision-spliced ANATOMY-V2 probes are excluded here as everywhere else in the paper.

Why we report two sample sizes. The 27 probes span only 11 pretraining branches, because several are different epochs of one training run. Probes within a branch share an encoder lineage, a seed and the test split, so treating them as 27 independent observations is pseudo-replication. Table 4 therefore reports each trend twice: once per checkpoint, and once after averaging within each branch. Where the two disagree, the branch-level figure is the one to believe.

Race. The black subgroup has the lowest AUC in every one of the 27 probes (n=431 black, n=251 asian, n=2318 white). At matched epoch 100 the black subgroup improves from 0.8325 under RANDOM to 0.8472 under INTENSITY ($+0.0147$), a larger absolute gain than the white subgroup's ($0.8741 \rightarrow 0.8853$). The max–min gap nonetheless widens slightly, because the asian subgroup — already the highest — gains most ($0.9033 \rightarrow 0.9189$). A widening max–min gap with universally rising subgroup AUCs is not evidence of harm, and we do not present it as such. The checkpoint-level correlation between aggregate AUC and racial gap is $\rho = +0.487$ (p=0.0100), which fails BH correction (q=0.0232) and vanishes at branch level (ρ=+0.518, p=0.1025). We therefore explicitly do *not* claim that more accurate models are less fair.

Disease severity. FairVision's label is defined by visual-field mean deviation (md): every test volume with md ≤ -2 is positive. Severity therefore cannot be used as an ordinary subgroup axis,

Table 4: Subgroup gap trends. "gap range" is the max–min subgroup AUC gap observed across the 27 valid probes. q is Benjamini–Hochberg across the seven attributes tested, which is necessary because seven trends were examined. Only the sex gap and the severity gap survive correction, and both *narrow* as aggregate AUC rises.

			per checkpoint (n=27)			per branch (n=11)	
attribute	worst group	gap range	ρ	p	q	ρ	p
sex	female (27/27)	0.0270–0.0396	-0.739	<0.0001	<0.0001	-0.891	0.0002
race	black (27/27)	0.0433–0.0935	+0.487	0.0100	0.0232	+0.518	0.1025
ethnicity	hispanic (27/27)	0.0203–0.0661	+0.151	0.4509	0.4509	+0.409	0.2115
language	other languages (22/27)	0.0518–0.0805	+0.184	0.3589	0.4187	+0.400	0.2229
marital status	single (27/27)	0.0349–0.0654	-0.380	0.0503	0.0881	-0.373	0.2589
age	<60 (26/27)	0.0474–0.0704	+0.300	0.1279	0.1791	+0.236	0.4841
disease severity	mild (-6,-2] (27/27)	0.1266–0.1400	-0.551	0.0029	0.0102	-0.618	0.0426

because within-bin AUC is undefined — a trap that silently yields nothing under naive stratification. Instead each positive stratum is scored against the shared pool of all 1534 negatives.

Table 5: Severity-stratified AUC at matched epoch 100. Each positive stratum is scored against the shared pool of all 1534 negatives. Every stratum improves, and mild disease improves most.

stratum	n_+	RANDOM	ENVELOPE	INTENSITY	Δ
severe (md ≤ -12)	334	0.9525	0.9559	0.9588	+0.0063
moderate ($-12 <$ md ≤ -6)	460	0.9030	0.9103	0.9131	+0.0102
mild ($-6 <$ md ≤ -2)	672	0.8164	0.8232	0.8301	**+0.0137**

The severe-to-mild gap is 0.1361 for the null and 0.1286 for the strongest policy, and stays within 0.0133 (0.1266–0.1400) across all 27 probes while aggregate AUC spans 0.8483–0.8855. Target placement raises the whole severity curve — mild disease most of all — without altering the ordering that makes early disease hard. We note that mild functional loss is harder to separate from normal OCT by construction, so this ordering is expected clinically; what the data add is that no masking policy we tested changes it.

Limitations. One dataset, one test split, reused throughout this programme's history. Because policies, checkpoints and analyses were chosen after repeated inspection of that split, the p-values here should be read as descriptive rather than as confirmatory inference. Race and ethnicity are self-reported categories inherited from FairVision's coding; at these subgroup sizes we cannot speak to intersectional groups. All figures are frozen mean-pool linear probes with AUC as the only metric; we report no subgroup calibration, no sensitivity at a fixed specificity, and no threshold-transfer analysis, so this audit should not be read as a complete fairness evaluation. Disparities could differ under fine-tuning or a stronger head.

D Excluded runs

Four probes of an earlier coverage arm are excluded from all analysis. They were trained with half-precision EMA targets and a different encoder-truncation policy from the arms they would be compared against, so their deficit is not attributable to masking. We report their existence for completeness and do not use them to support any claim. Similarly, the ANATOMY-V2 probes at epochs 75 and 92 follow a change of target precision partway through that arm's training; they are listed in Table 3 but excluded from the matched-epoch comparisons in Table 1.

E Occlusion attribution: where the classifier looks

The analyses in this appendix were run on the three *fine-tuned* probes (mean-pool, cross-attention pool, and a depth-1 attentive probe) that tie at test AUC ≈ 0.887, not on the frozen probes used for the arm comparison in the main text. They characterise what the downstream classifier uses, and they are reported here because two of the findings are cautionary rather than positive.

Method. Attribution is by *occlusion*, which is architecture-agnostic and causal. Gradient$\times$input is not used because two of the three probes are non-linear in the slice-token matrix, where gradient attribution is misleading. For slice-level attribution each slice token is zeroed in turn and Δlogit recorded; for window-level, seven consecutive slices; for patch-level, leave-one-out over the 256 patch tokens of a target slice.

A bimodal attribution curve that is an artefact, not anatomy. The population-averaged curve peaks at native slice ≈ 63 and ≈ 137 with a dip near 95. The tempting reading — that the model has discovered the superior and inferior optic disc rim — is **wrong**, and we report it because it is exactly the kind of claim a plausible-looking attribution figure invites. Three tests reject it. First, if the two peaks were bilateral anatomy then within a diseased eye they should co-occur, giving positive per-volume correlation between them; the observed correlation is slightly *negative* (-0.22, -0.07, -0.14 across the three probes). Second, clustering the per-volume curves into two groups yields clusters that are near-perfect *mirror images* along the slice axis ($\mathrm{corr}(c_1, \mathrm{flip}(c_2)) = 0.971$ and 0.988 for the two well-structured probes, against raw correlations of -0.124 and -0.478). Third, using that cluster label as pseudo-laterality and flipping one cluster collapses the bimodality asymmetrically. The signature is OD/OS axial storage: each eye carries its dominant signal on one side of the disc, right and left eyes are stored with opposite axial orientation, and population averaging manufactures an illusion of symmetry.

Errors are weaker signal, not different anatomy. Stratifying by outcome, false-negative curves are scaled-down true-positive curves with the same shape, and false positives likewise mirror true negatives. Attribution structure is also near-invariant to prediction confidence ($|r| \leq 0.25$ between peak contribution magnitude and $|\mathrm{logit}|$). The error rate at threshold 0.5 therefore reflects signal-strength saturation on hard cases, not the model attending to the wrong structures.

Why this is in a masking paper. Two of these results bear directly on the main claim. The classifier's evidence is spread across the middle two-thirds of the volume and across most patches within a slice, rather than concentrated on a compact anatomical target. A masking policy that sharpens targets onto a precise anatomical boundary is therefore optimising for a spatial prior the downstream task does not appear to need, which is consistent with Section 5.2 finding no relationship between anatomical specificity and downstream AUC. The OD/OS result is a methodological caution of the same family as our precision and epoch-matching checks: a figure that looks anatomically meaningful can be an artefact of data storage, and we would rather report that than let it stand.

F Broader impact and ethics

This study uses a public, de-identified retinal dataset (FairVision) under its release terms and involves no new data collection and no human subjects. No model reported here is clinically validated or intended for deployment: the frozen-probe AUCs are research measurements on one dataset and one task, and are not evidence of clinical utility.

The subgroup analysis in Section 5.3 is a caution, not a contribution. Every masking policy we tested leaves the same groups worst-served, and the best-performing policy does not close any gap. We would regard deployment of any encoder reported here, without a dedicated and prospective fairness evaluation on the target population, as inappropriate. We also stress that the audit is AUC-only: it contains no subgroup calibration, no sensitivity at a fixed clinical specificity, no predictive-value analysis and no intersectional breakdown, so it should not be mistaken for a complete equity assessment.

Releasing pretrained weights carries the usual dual-use consideration that they may be applied to populations unlike the training cohort. We document the cohort composition (Appendix C) so that such a mismatch is detectable by anyone reusing the weights. The demographic attributes are self-reported categories inherited from the dataset's coding; they are used only for the subgroup audit and are never model inputs.

G Published ablations on informed versus random masking

Our result should be read against the following published comparisons, which we collected while positioning this work. The pattern is that random masking is a strong baseline and informed masking wins only under some objectives and protocols; the deficit is largest under frozen evaluation.

Table 6: Published ablations comparing a random or simple baseline against an informed or structured masking scheme. Numbers are as reported by the cited papers; metrics differ across rows and are not comparable between rows.

study	baseline	informed variant	interpretation
MAE [He et al., 2022]	random-75: 84.9 FT / 73.5 lin	block-75: 82.8 / 63.9; grid-75: 84.0 / 66.0	random decisively better
SimMIM [Xie et al., 2022]	83.0 top-1	best block 82.7; square 82.6	random modestly better
SemMAE [Li et al., 2022]	random 66.8 lin	within-part 66.5; whole-part 52.9; adaptive 68.7	naive semantics hurt; scheduled mantics help
SemMAE part quality [Li et al., 2022]	no parts 63.7	raw parts 63.6; refined 65.0	an imperfect semantic model a nothing
AutoMAE [Chen et al., 2023a]	MAE 83.26 FT	AutoMAE 83.32	effectively tied
HPM [Wang et al., 2023b]	random 82.49	moderate 82.95; hardest-only 81.40	benefit only with retained rand ness
AttMask [Kakogeorgiou et al., 2022]	random 43.4 k-NN	high 43.5; hint 43.6	nearly indistinguishable
AttG-MAE [Sick et al., 2025]	MAE 78.5 @10% labels	guided 78.1	guidance loses in this protocol
SSiT [Huang et al., 2024]	no saliency 79.48 κ	25% mask 77.53; 50% 79.97	dataset-dependent, non-monotoni

Read together, these say that the *direction* of our finding is not new. What we add is a controlled measurement in a setting where the guidance comes from a trained medical segmentation model rather than from attention or difficulty, and where the comparison is run from a shared checkpoint with the masking policy as the only difference.

H Full paired-contrast table

I Full fp32 re-probe

The RANDOM, INTENSITY and ENVELOPE long-horizon probes were originally fitted under the harness default, which enables autocast. To remove any doubt that the headline effect is a precision artifact, we re-ran the complete probe pipeline for those arms with autocast disabled, so that feature extraction, head optimisation and test evaluation are all fp32, under the exact protocol already used by the ANATOMY and COVER arms. The encoder checkpoint is SHA-256 hashed before and after each run and verified unchanged, so no result here can be contaminated by an accidental encoder update.

fp32 re-probe table pending†

J Fine-tuning narrows but does not erase the gap

Under full fine-tuning rather than a frozen probe, INTENSITY reaches 0.8947 (mean-pool head) against RANDOM's 0.8868, a gap of +0.0079 compared with +0.0109 frozen. Taking each arm's best head instead, the gap is +0.0069. The pretraining difference is partially, but not entirely, absorbed by supervised adaptation — consistent with the masking policy shaping the representation rather than merely the optimisation start point. These runs used a different head and schedule from the frozen probes and are never pooled with them.

K Fine-tuned results

Table 7: Primary contrasts: matched epoch *and* matched precision. Δ is a paired difference on identical test cases; the interval is a 10,000-resample paired bootstrap and p is a DeLong test for correlated ROC curves. q is Benjamini–Hochberg over these nine pre-specified contrasts. Every comparison against the null excludes zero.

contrast	epoch	Δ AUC	95% CI	p	q
ENVELOPE − RANDOM	50	+0.0120	[+0.0068, +0.0173]	<0.0001	<0.0001
ENVELOPE − RANDOM	75	+0.0080	[+0.0028, +0.0132]	0.0025	0.0045
ENVELOPE − RANDOM	100	+0.0062	[+0.0010, +0.0113]	0.0199	0.0299
INTENSITY − RANDOM	50	+0.0099	[+0.0051, +0.0149]	<0.0001	0.0001
INTENSITY − RANDOM	75	+0.0113	[+0.0063, +0.0164]	<0.0001	<0.0001
INTENSITY − RANDOM	100	+0.0109	[+0.0057, +0.0160]	<0.0001	<0.0001
INTENSITY − ENVELOPE	50	-0.0020	[-0.0069, +0.0029]	0.4114	0.4114
INTENSITY − ENVELOPE	75	+0.0033	[-0.0013, +0.0079]	0.1491	0.1677
INTENSITY − ENVELOPE	100	+0.0047	[+0.0004, +0.0091]	0.0294	0.0378

Table 8: Full fine-tuning, same test split. Reported separately from frozen probes and never pooled with them.

policy	head	test AUC
INTENSITY	mean-pool	0.894668
INTENSITY	cross-attention	0.893717
INTENSITY	attentive	0.890100
RANDOM	attentive	0.887763
RANDOM	cross-attention	0.887178
RANDOM	mean-pool	0.886756

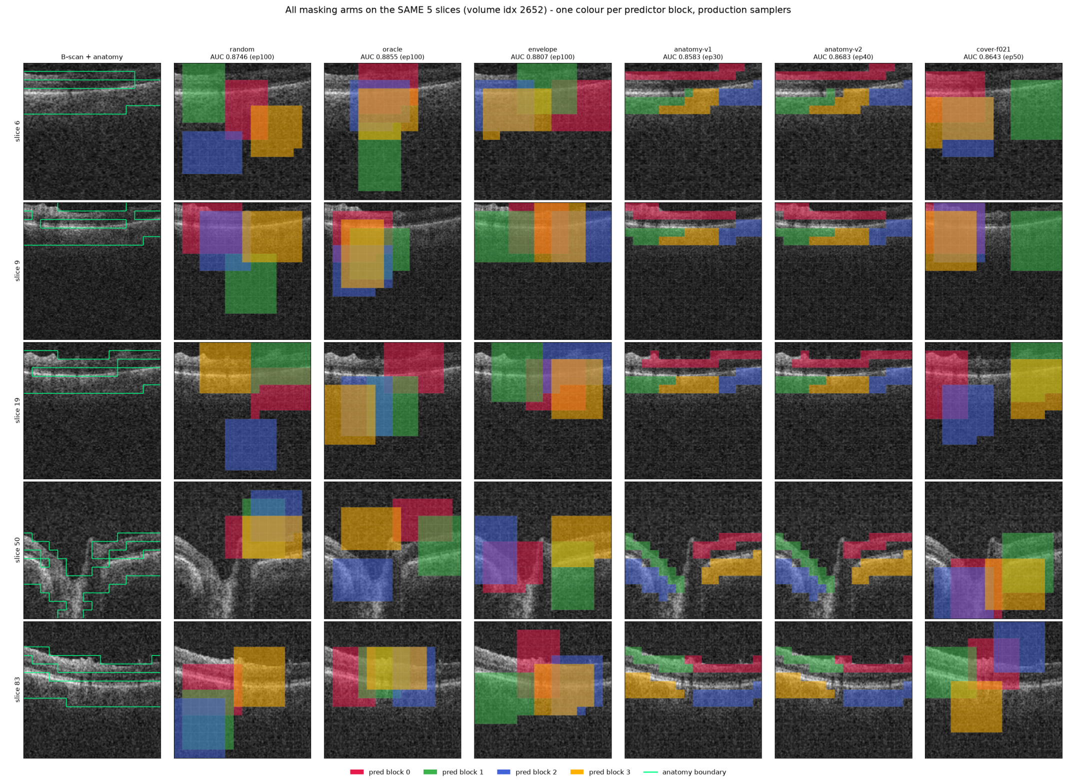

Figure 4: All six masking policies (columns) on five B-scans (rows) spanning one volume, drawn with the production samplers. Leftmost column is the unmasked B-scan. Each predictor target block has its own colour; the green contour is the anatomy reference. The random, envelope, and cover arms place axis-aligned rectangles; the anatomy arms grow cell-wise targets that follow the retinal band and therefore change shape from slice to slice.

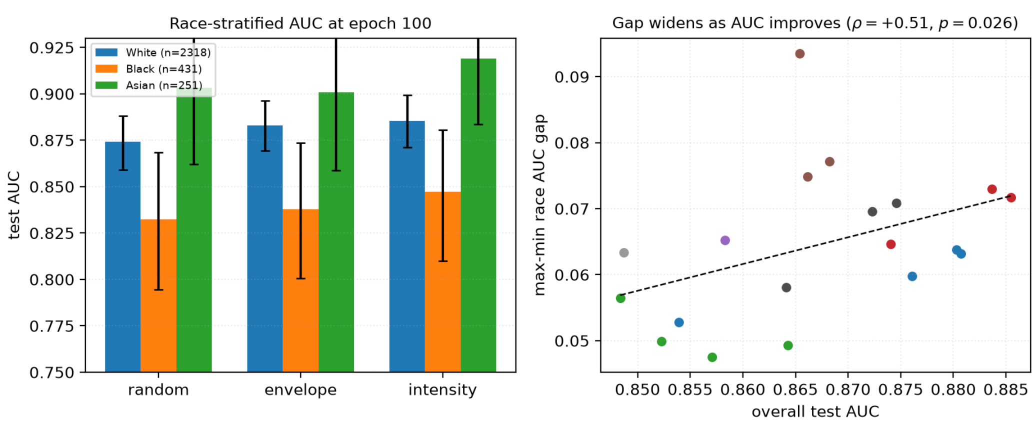

Figure 5: **Left:** race-stratified AUC at epoch 100 with bootstrap intervals. The black subgroup is lowest under every policy, but note that *every* group improves from RANDOM to INTENSITY. **Right:** max–min race gap against aggregate AUC across 27 checkpoints. The fitted slope is positive but does not survive multiplicity correction or branch-level aggregation (Table 4), so we draw no trend conclusion from it.

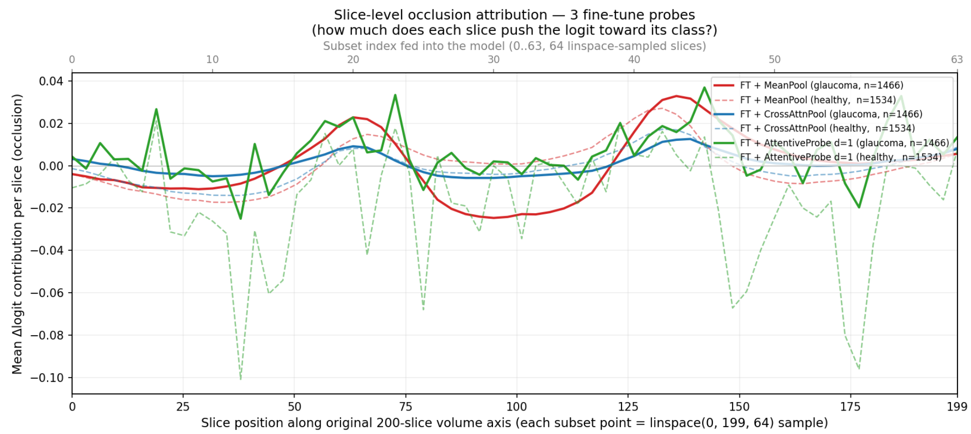

Figure 6: Single-slice occlusion attribution against native slice position, averaged over all 3000 test volumes, for all three fine-tuned probes. Despite spanning zero to 7.17M probe parameters they converge on the same structure (mean-pool vs cross-attention pool $r{=}0.94$). The y-axis is *signed* Δlogit: peaks are slices whose removal lowers the logit, the central dip is slices whose removal raises it, and only slices near zero are genuinely unused.

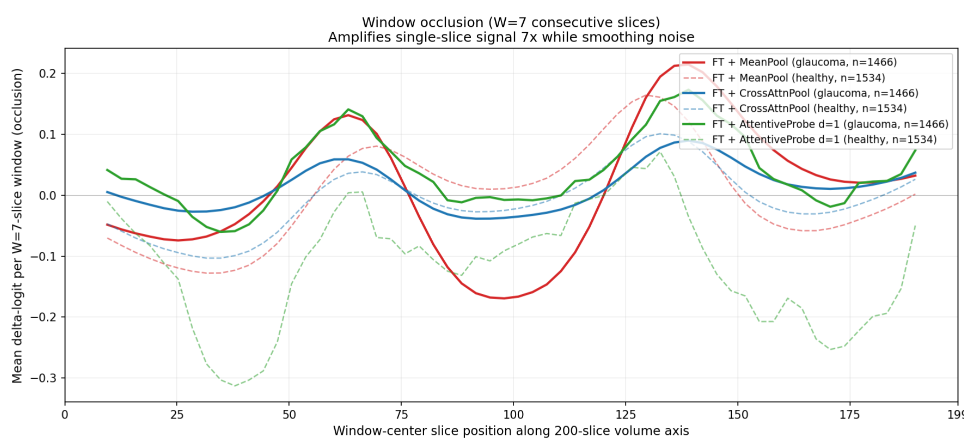

Figure 7: Window occlusion ($W{=}7$) amplifies the same signal roughly sevenfold and suppresses single-volume noise. For mean-pooled models, single-slice occlusion systematically underestimates attributed signal; the windowed variant recovers about $25\times$ more. We recommend it as the default primitive for this class of model.

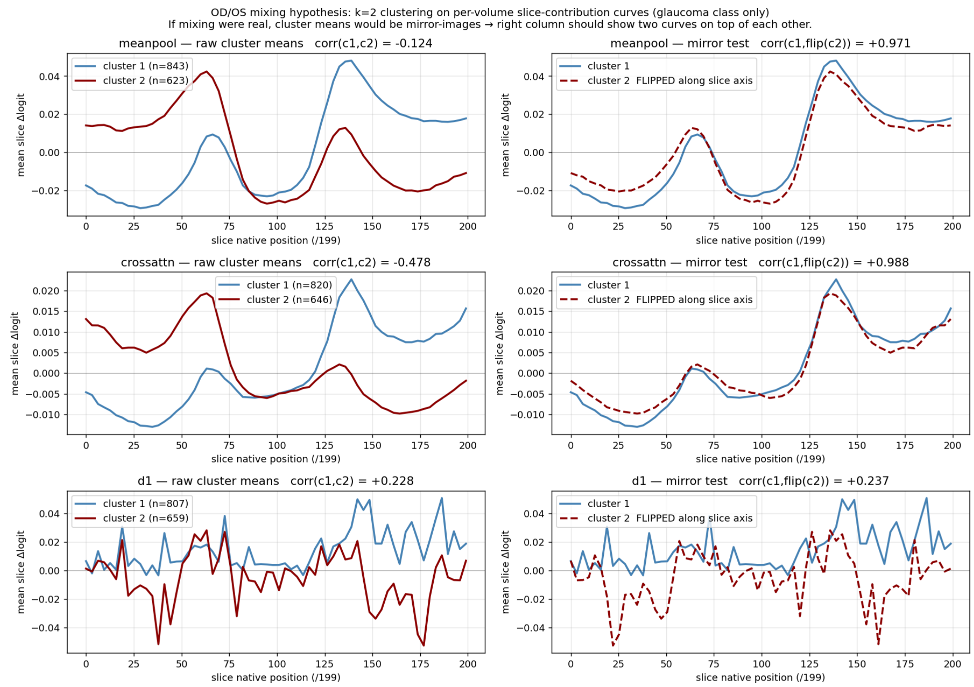

Figure 8: The two-peak structure is an OD/OS storage artefact. Clustering per-volume attribution curves into two groups returns near-perfect mirror images of one another, which is the direct signature of laterality flipping rather than bilateral anatomy.

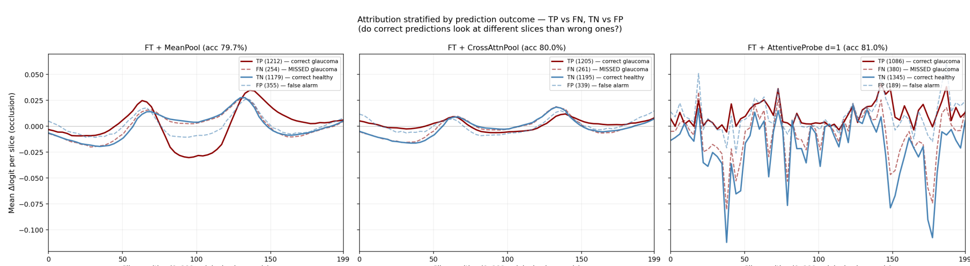

Figure 9: Attribution stratified by TP / FN / TN / FP. Errors reproduce the correct pattern at lower amplitude rather than selecting different anatomy.

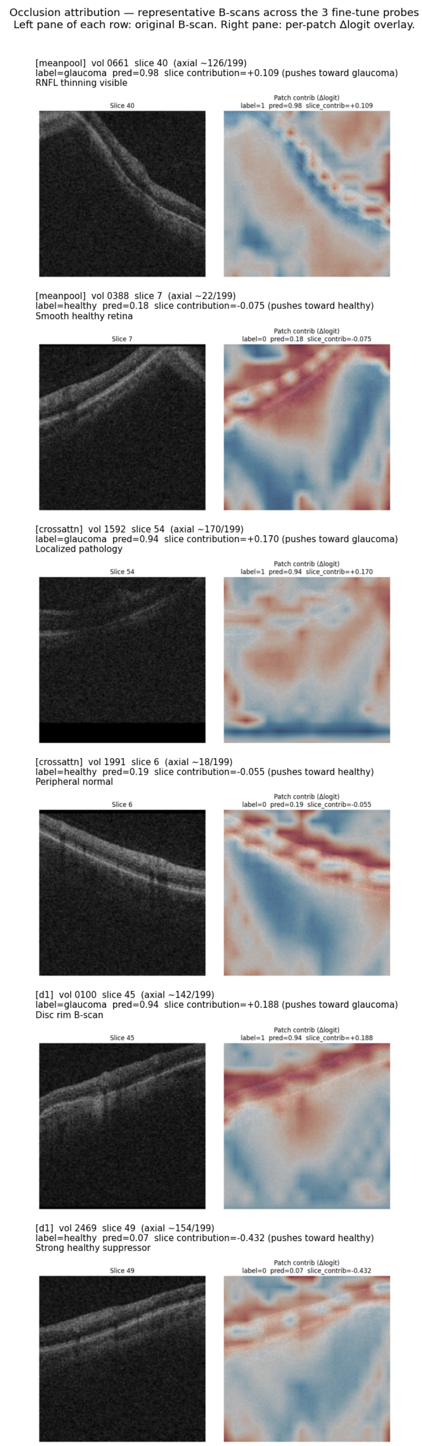

Figure 10: Per-patch occlusion heat maps on individual B-scans, drawn on a shared scale with the slice mean subtracted so that maps are comparable across volumes. Between 84% and 91% of patches have a 95% bootstrap interval excluding zero on the glaucoma class (B=500), so attribution is broadly distributed rather than concentrated in a few patches. Cross-probe agreement is strong at slice level (r=0.94) and moderate at patch level (per-volume $r \approx 0.35$–0.48).

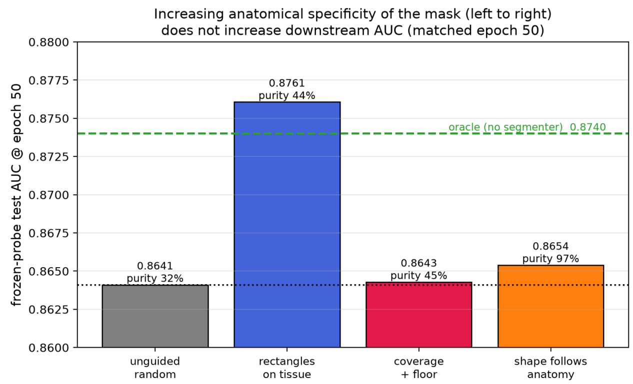

Figure 11: The matched-epoch-50 result arranged as a ladder of increasing anatomical specificity. Companion to Figure 3.

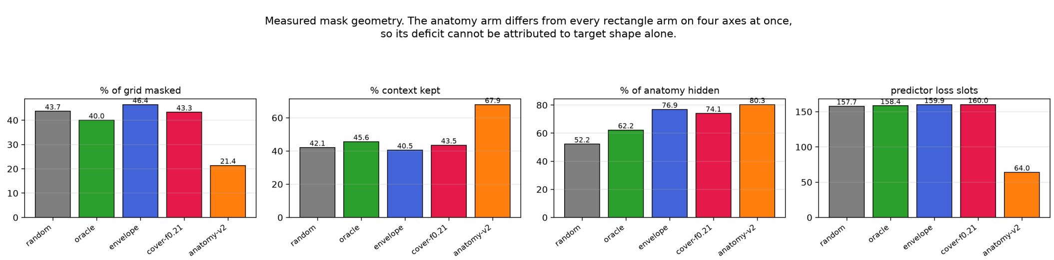

Figure 12: Measured mask geometry per policy, 600 held-out slices. The four rectangle policies are closely matched on every axis, so their comparison is near single-variable. The anatomy policy diverges on all four simultaneously — half the masking ratio, $1.6\times$ the context, $0.4\times$ the predictor loss slots — which is why we decline to attribute its deficit to target shape.